7 (a) (i) photons and electrons undergo a <u>one-to-one</u> interaction. B1

(ii) photons could have interacted with electrons below surface; energy is used to take electron to the surface. B1

or

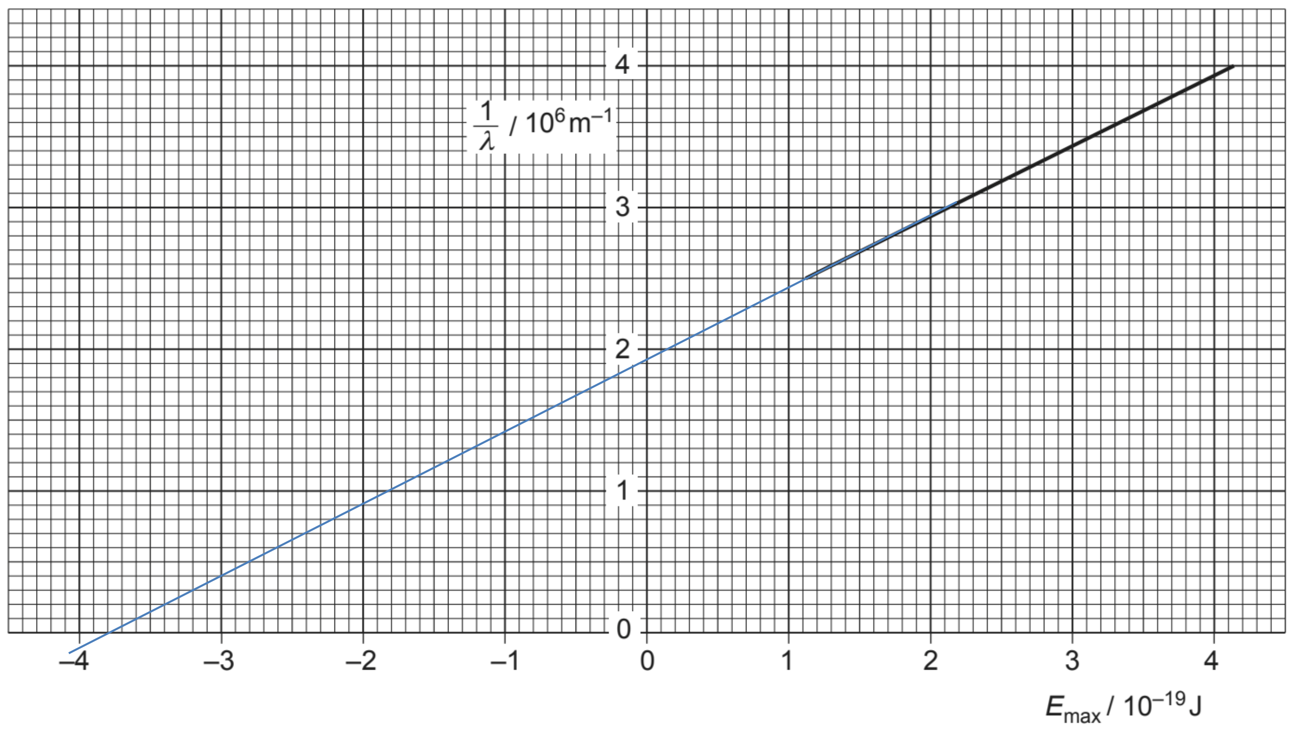
some electrons are more tightly bound to nuclei than others

(iii) 1. $\phi = -E_{max}$ when $\frac{1}{\lambda} = 0$ (use of the x-intercept) B1
Allow range $(3.8 \pm 0.2) \times 10^{-19}$ J

(allow use of substitution)

2. Use of gradient $= \frac{1}{hc}$ C1
Allow range of gradient $(4.8 - 5.2\ ) \times 10^{24}$ B1
Allow range of *h* $(6.4 - 6.9)\ \times 10^{-34}$ J s

alternative
(allow use of substitution of points) C1
Allow range of *h* $(6.4 - 6.9)\ \times 10^{-34}$ J s A1



(iv) increase intensity only increases the rate / number per unit time of the emitted electrons B1
E_{max} depends on frequency of photons and work function of metal only. B1

Hence, no change.

(b)	**(i)**	energy difference $\Delta E = \frac{hc}{\lambda} = \frac{6.63\times10^{-34}\times3.0\times10^{8}}{663\times10^{-19}} = 3.0 \times 10^{-19}$	M1
		So energy levels associated with the transition is between n = 2 and n = 3	B1
		Since dark line is observed, n = 2 to n = 3	A1
	(ii)	$21.8 \times 1 = 21.8$ $5.4 \times 2^2 = 21.6$ $2.4 \times 3^2 = 21.6$	
		1 mark for working to find constant 1 mark for comparing the any pair of values	B2

8 **(a)** **(i)** Similarity: B1